

Wellington Santos

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Artificial Intelligence Engineer with **8+** years across sports commerce, retail e-commerce, and healthcare IT, building scalable AI solutions and intelligent automation workflows. Expert in LLM integration, NLP pipelines, and enterprise AI deployment using Python, OpenAI APIs, and cloud infrastructure. Proven ability to translate complex business challenges into production-grade AI systems delivering measurable operational impact.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Generative AI: LangChain, OpenAI API, LlamaIndex, Anthropic API, Hugging Face, LangGraph, Pinecone, ChromaDB, Weaviate, FAISS, MLflow, LangSmith

Data & ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Spark, Airflow, dbt

Backend: FastAPI, Express, NestJS, Django, Spring Boot, Flask, RabbitMQ, Kafka

Frontend: React, Next.js, Vue, Angular, Redux, Tailwind CSS

Database: PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB, MySQL

Testing: Jest, Pytest, Playwright, Cypress, Vitest

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Lambda, CloudFront, EC2, RDS, S3

Others: RESTful API, GraphQL, gRPC, WebSocket, JSON, Jira, Confluence, Git, GitHub, Grafana, Prometheus, Datadog, Agile, Scrum

EXPERIENCE

Fanatics
Senior Software Engineer

Oct 2023–Present

- Led the development of an AI-powered personalization engine for a high-scale sports fan ecosystem, integrating **Python**, **LangChain**, and **OpenAI API** to deliver real-time content recommendations that increased engagement by **28%** across collectibles and gaming modules.
- Drove the implementation of NLP-based chatbot infrastructure using **Anthropic API** and **LlamaIndex** for semantic search over player statistics and betting odds, enabling instant fan queries while reducing customer support volume by **40%**.
- Built end-to-end **FastAPI** microservices for AI agent orchestration, connecting **LangGraph** agents with Redis-backed session management to coordinate multi-step betting workflow automation across the platform.

- Architected a **Pinecone**-powered embeddings pipeline that indexed 500k+ sports content items, enabling vector similarity search within **Kafka**-driven event streams for real-time personalization at peak traffic.
- Collaborated with product and analytics teams to establish MLflow monitoring dashboards tracking inference latency, token usage, and cost-per-prediction, optimizing model deployments to reduce LLM API spend by **22%**.
- Implemented comprehensive error handling and fallback logic across all AI services to gracefully handle LLM timeouts and rate limits, ensuring 99.2% uptime for critical recommendation flows.
- Mentored two junior engineers on LLM best practices, prompt optimization, and responsible AI governance, establishing team standards for input validation and output filtering.

Walmart

Sep 2022–Oct 2023

Senior Software Engineer

- Spearheaded a healthcare AI initiative by building intelligent document summarization workflows using **Python**, **OpenAI API**, and **FastAPI** to automatically process pharmacy claims and medical records, cutting manual review time by **35%** across 12 distribution centers.
- Designed and deployed NLP classification models with **Hugging Face** transformers to categorize incoming customer support tickets for retail, e-commerce, and financial services, routing them to specialized teams and improving resolution time by **18%**.
- Engineered a **LangChain**-based agent framework that integrated with Walmart's internal CRM and inventory APIs, enabling cross-functional teams to query multi-source data through natural language conversations without writing SQL.
- Developed embedding-based semantic search using **Weaviate** vector database to power intelligent product recommendations across Walmart.com, connecting customer queries to catalog data and improving click-through rates by **12%**.
- Collaborated with compliance teams to establish AI governance policies for regulated healthcare workflows, implementing **LangSmith** tracing for prompt auditing and output validation to meet HIPAA standards.
- Optimized **Anthropic API** token consumption for large-batch document processing by implementing intelligent chunking and caching strategies with **Redis**, reducing monthly AI infrastructure costs by **31%**.

E-POT

Jun 2018–July 2022

Full Stack Engineer

- Built a healthcare AI solution leveraging **Python** NLP frameworks and **LangChain** to extract clinical entities from unstructured medical notes, enabling automated data extraction for revenue cycle management and reducing manual chart review by **45%**.
- Developed **FastAPI**-based microservices integrating legacy healthcare systems with modern AI pipelines, connecting **PostgreSQL** databases to **OpenAI API** for clinical decision support workflows across 6 hospital networks.
- Implemented ETL data pipelines using **Airflow** and **Spark** to prepare medical datasets for machine learning, combining structured claims data from **MongoDB** with clinical narratives to train **scikit-learn** classification models.

- Created real-time **React** dashboards using **TypeScript** and **Redux** to visualize AI model performance metrics and predictions, giving clinicians visibility into model confidence scores and recommendation explanations.
- Designed **Docker**-based deployment architecture with **Kubernetes** orchestration for reproducible AI model serving, enabling seamless promotion of trained models from development to production across AWS and on-premises environments.
- Established CI/CD pipelines using **GitHub Actions** with automated retraining triggers, ensuring models stayed current with evolving medical data while maintaining audit trails for compliance documentation.
- Mentored three team members on full-stack AI development, covering model evaluation techniques, API integration patterns, and observability best practices using **Prometheus** and **Grafana** for production monitoring.

EDUCATION

University of Tokyo
Bachelor of Science in Computer Science

Apr 2014–Mar 2018